

Chandrika Sethumadhavan

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PROFILE SUMMARY

- M.Sc. Embedded Systems Engineering (Uni Freiburg, GPA 1.9); research focus on predictive ML for physical systems, time-series forecasting, scenario simulation, and multi-objective prescriptive optimization (NSGA-II). Published SSRN 2025.
- Agentic AI in practice: built a full RAG pipeline with LLM-API integration (OpenAI) and agentic document generation workflow, end-to-end product delivered at Black Forest Hackathon 2025.
- Production ML at Fraunhofer ISE: YOLOv8 inference pipeline (94.4% accuracy) for live industrial quality inspection. AWS MLA-C01 certified with hands-on MLOps, ONNX export, and INT8 quantization.
- Python (strong), PyTorch, TensorFlow, scikit-learn, Docker, CI/CD. English C2, German B1.

EDUCATION

University of Freiburg Germany
M.Sc. Embedded Systems Engineering 2023 to present
GPA: 1.9 — Focus: ML/MLOps, predictive modelling, physical simulation, agentic AI systems

Anna University India
B.E. Electrical & Electronics Engineering 2019-2023
GPA: 1.4

TECHNICAL SKILLS

Predictive & Prescriptive ML: Time-series forecasting (LSTM, GRU), transfer learning and domain adaptation, physics-informed hybrid models, multi-objective optimization (NSGA-II), scenario modelling, Pareto-front analysis, AutoML (zero-cost proxy NAS, Optuna)

Agentic AI & LLMs: RAG (Retrieval-Augmented Generation), LLM-API integration (OpenAI), prompt engineering, agentic workflow design, LLMOps, responsible AI practices

ML Frameworks: PyTorch, TensorFlow (+Lite), scikit-learn, YOLOv8, Optuna, Edge Impulse

MLOps & Deployment: AWS (SageMaker, EC2, S3, MLA-C01 certified), Docker, GitHub/GitLab, CI/CD pipelines, FastAPI, Flask, ONNX, INT8 quantization, model monitoring and drift detection

Data & Pipelines: pandas, NumPy, SciPy, SQL, MQTT, InfluxDB (Cloud, time-series), Grafana, REST APIs

Languages: Python (strong), C, C++, SQL

WORK EXPERIENCE

Master Thesis, IMTEK, University of Freiburg Germany
Research Scientist (Thesis) Ongoing

- Built predictive ML models (LSTM/GRU with transfer learning) for time-series forecasting of actuator system dynamics under sparse data. Pipeline pre-trained on source domain, fine-tuned to target domain.
- Simulation-based scenario modelling with COMSOL Multiphysics (FEM) to map feasibility regions; NSGA-II multi-objective optimization for prescriptive identification of optimal operating windows.
- Benchmarked physics-only, data-only, and transfer learning variants on RMSE, MAE, and R^2 . Results published on SSRN (DOI: 10.2139/ssrn.6504441, 2025).

Working Student at Fraunhofer ISE, Freiburg Germany
Image Processing & ML Dec 2023-Jun 2024

- Developed and deployed a YOLOv8 real-time inference pipeline for industrial production-line quality inspection on solar cells; **94.4%** accuracy on live manufacturing data.

- Statistical analysis with OpenCV, NumPy, SciPy; integrated pipeline with industrial 3D laser microscope (LEXT) for automated quality control.

Working Student at ECOSENSE, IMTEK, Uni Freiburg

Germany

Embedded Systems & Data Pipelines

Sep 2024-Nov 2025

- Built a Python data ingestion pipeline (MQTT → InfluxDB Cloud → Grafana) for real-time environmental sensor monitoring with live dashboard visualization.

KEY PROJECTS

Agentic AI System: GenAI Documentation Assistant, Black Forest Hackathon 2025 GitHub

- Built a full agentic AI workflow: Vision Transformer (EndoViT/PyTorch) for image analysis, RAG pipeline for knowledge-grounded context retrieval, and OpenAI LLM for structured automated report generation.
- Python REST API backend with React/TypeScript frontend. Full GenAI product delivered end-to-end within 48 hours; responsible AI practices applied throughout via RAG grounding and structured output validation.

Carbon-Aware AutoML Pipeline with ONNX Deployment

GitHub

- End-to-end ML lifecycle: zero-cost proxy NAS, Optuna HPO, CodeCarbon emissions tracking, ONNX export with INT8 quantization, and multi-backend benchmarking (PyTorch CPU, ORT CPU, ORT CUDA).
- **1.87×** CPU and **8.54×** GPU speedup; **74.8%** model size reduction; accuracy up to **97.0%** across five datasets.

CERTIFICATIONS

AWS Certified Machine Learning Engineer, Associate (MLA-C01)

Amazon Web Services

End-to-end ML pipelines, SageMaker, model monitoring, drift detection, responsible AI and LLMOps best practices.

Tiny Machine Learning (TinyML), HarvardX (edX)

Harvard University

PUBLICATIONS

Kinetic Modeling and Quantification of Oxygen Release from Anthracene Endoperoxides under Confined Conditions, SSRN, DOI: 10.2139/ssrn.6504441, 2025.

Autonomous Delivery Vehicle Using Raspberry Pi and Computer Vision, Springer, PC-CDA'23, 2023.

LANGUAGES

English: C2 — German: B1